

1. Consider the following proposition, which we could label as $P(n)$.

If $5n + 3$ is even, then n is odd.

Prove by strong induction that for every positive integer n , proposition $P(n)$ is true.

Note that I don't think this holds as k might be getting redefined

Proof :

By strong induction, assume that $P(k)$ is true for $1 \leq k \leq n - 1$

The base case of $n = 2$ ($n = 1$ produces an odd value for $P(n)$, so it is not a base case) holds as $P(2) = 5(2) + 3 = 13$ which is odd, and 2 is even.

By contrapositive, assume $k + 1 = 2b$ where b is an integer

$k = 2b - 1$, which is a contradiction thus proving the inductive step through contrapositive. ■

2. Prove by strong induction that for every positive integer n the following holds:

$$\sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}$$

proof:

By strong induction, assume:

$$\sum_{k=1}^{n-1} k(k+1) = \frac{n(n+1)(n-1)}{3}$$

The base case $n = 1$ holds:

$$\sum_{k=1}^1 k(k+1) = 1(1+1) = 2 = \frac{1(1+1)(1+2)}{3}$$

By substituting the last term of the summation:

$$\sum_{k=1}^n k(k+1) = \frac{n(n-1)(n+1)}{3} + n(n+1)$$

The inductive hypothesis is proved below:

$$\frac{n(n-1)(n+1)}{3} + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$n(n-1)(n+1) + 3n(n+1) = n(n+1)(n+2)$$

$$n^3 - n + 3n^2 + 3n = n^3 + 3n + 2n$$

$$n^3 + 3n^2 + 2n = n^3 + 3n^2 + 2n$$

■